Lecture 9

The arithmetic of characters

Objectives. *Character values are sums of roots of unity, so they are* algebraic integers. *This lecture turns that arithmetic into theorems about groups: every irreducible complex representation has degree dividing $|G|$. Burnside's theorem connects group structure with representation theory: groups of order $p^a q^b$ are solvable. Last, the Frobenius–Schur indicator decides which invariant bilinear form an irreducible representation carries.*

By the end of this lecture you will be able to:

- Prove closure of algebraic integers under addition and multiplication, and integrality of character values and central characters; deduce that every irreducible degree divides $|G|$.
- Prove that an average of roots of unity that is an algebraic integer is 0 or the roots of unity averaged are all equal.
- Prove that a group with a conjugacy class of prime-power size greater than 1 is not simple, and that groups of order $p^a q^b$, for distinct primes p, q and $a, b \geq 0$, are solvable.
- Compute the Frobenius–Schur indicator of an irreducible representation, and decide from it which invariant bilinear forms it carries.

9.1 Algebraic integers

Throughout, the group G is finite, and representations are complex and finite-dimensional.

Definition 9.1 (Algebraic integer)**.** A complex number α is an *algebraic integer* if $h(\alpha) = 0$ for a monic polynomial

$$h(t) = t^n + c_{n-1}t^{n-1} + \cdots + c_1 t + c_0 \qquad (n \geq 1)$$

with integer coefficients c_i.

Every integer a is one, as a root of $t - a$, and so is every root of unity, a root of some $t^m - 1$.

The *companion matrix* of a monic h of degree n, with any complex coefficients, is the $n \times n$ matrix C with entries 1 just above the diagonal, last row $(-c_0, \ldots, -c_{n-1})$, and 0 elsewhere. If $h(\alpha) = 0$, then $v = (1, \alpha, \ldots, \alpha^{n-1})^{\mathsf{T}}$ satisfies $Cv = \alpha v$: for $i < n$,

$$(Cv)_i = \alpha^i = \alpha v_i,$$

and

$$(Cv)_n = -c_0 - c_1\alpha - \cdots - c_{n-1}\alpha^{n-1} = \alpha^n = \alpha v_n.$$

Lemma 9.2 (Integer matrices)**.** *α is an algebraic integer if and only if it is an eigenvalue of a square matrix with integer entries.*

Proof. If $Av = \alpha v$ with $v \neq 0$, then α is a root of $\det(t\mathbb{1} - A)$, which is monic with integer coefficients, by expanding the determinant. Conversely, if $h(\alpha) = 0$, then α is an eigenvalue of the companion matrix of h, as $v \neq 0$. This companion matrix has integer entries because h has integer coefficients. ∎

Theorem 9.3 (Algebraic integers)**.**

(i) If α and β are algebraic integers, so are $\alpha + \beta$ and $\alpha\beta$.

(ii) A rational algebraic integer is an integer.

(iii) For every representation (ρ, V) and $g \in G$, $\chi_V(g)$ and $\overline{\chi_V(g)}$ are algebraic integers.

Proof. (i) By Lemma 9.2, $Av = \alpha v$ and $Bw = \beta w$ for integer matrices A and B and non-zero vectors v and w. Then $v \otimes w \neq 0$, and

$$(A \otimes \mathbb{1} + \mathbb{1} \otimes B)(v \otimes w) = (\alpha + \beta)(v \otimes w),$$
$$(A \otimes B)(v \otimes w) = \alpha\beta\,(v \otimes w).$$

Both matrices have integer entries; apply Lemma 9.2 again.

(ii) Let $\alpha = a/b$ with integers a and $b \geq 1$ coprime, and $h(\alpha) = 0$ as in Definition 9.1. Multiplying by b^n,

$$a^n = -\big(c_{n-1}a^{n-1}b + \cdots + c_1 a b^{n-1} + c_0 b^n\big),$$

so b divides a^n. A prime dividing b would divide a; so $b = 1$.

(iii) Put $d = \dim V$. By Corollary 5.17(ii), $\rho(g)$ is diagonalisable with eigenvalues $\zeta_1, \ldots, \zeta_d$, roots of unity, so $\chi_V(g) = \zeta_1 + \cdots + \zeta_d$. The conjugate $\overline{\zeta_i} = \zeta_i^{-1}$ is a root of unity too. Apply (i). If $V = 0$, both character values are 0, an algebraic integer as a root of t. ∎

For example, $\sqrt{2}$ is an algebraic integer because it is a root of $t^2 - 2$. The rational number $\frac{1}{2}$ is not an algebraic integer: it is not an integer, so the rational-integer criterion (Theorem 9.3(ii)) rules it out.

9.2 Central characters and $\dim U \mid |G|$

Let the classes of G be $\mathcal{K}_1 = \{e\}, \mathcal{K}_2, \ldots, \mathcal{K}_r$, with class sums $\widehat{\mathcal{K}_i} = \sum_{x \in \mathcal{K}_i} x$ and chosen $g_k \in \mathcal{K}_k$. By Theorem 7.11, the class sums are a basis of $Z(\mathbb{C}[G])$, and central elements act on each irreducible by scalars.

Definition 9.4 (Central character)**.** For $U \in \mathrm{Irr}(G)$, the *central character* $\omega_U : Z(\mathbb{C}[G]) \to \mathbb{C}$ is defined by $\rho_U(z) = \omega_U(z)\mathbb{1}$. For a class $\mathcal{K}$, write $\omega_U(\mathcal{K}) = \omega_U(\widehat{\mathcal{K}})$.

Lemma 9.5 (Central characters and class sums)**.** *Let $U \in \mathrm{Irr}(G)$.*

(i) ω_U is an algebra homomorphism, and for $g \in \mathcal{K}$,

$$\omega_U(\mathcal{K}) = \frac{|\mathcal{K}|\ \chi_U(g)}{\dim U}.$$

(ii) $\widehat{\mathcal{K}_i}\widehat{\mathcal{K}_j} = \sum_k a_{ijk}\widehat{\mathcal{K}_k}$, where a_{ijk} is the number of pairs $(x, y) \in \mathcal{K}_i \times \mathcal{K}_j$ with $xy = g_k$.

Proof. (i) ρ_U is an algebra homomorphism (Proposition 7.5(i)), so ω_U is one too. Take the trace of $\rho_U(\widehat{\mathcal{K}}) = \omega_U(\mathcal{K})\mathbb{1}$; as χ_U is constant on $\mathcal{K}$,

$$\omega_U(\mathcal{K}) \dim U = \sum_{x \in \mathcal{K}} \chi_U(x) = |\mathcal{K}|\ \chi_U(g),$$

and $\dim U \geq 1$.

(ii) The product is central, so it is a combination $\sum_k a_{ijk}\widehat{\mathcal{K}_k}$ (Theorem 7.11(i)). Its coefficient of g_k is a_{ijk}, as g_k lies in $\mathcal{K}_k$ only; in $\sum_{x\in\mathcal{K}_i,\,y\in\mathcal{K}_j} xy$ it is the number of pairs with $xy = g_k$. ∎

Theorem 9.6 (Integrality and divisibility). *Let $U \in \mathrm{Irr}(G)$.*

(i) Each $\omega_U(\mathcal{K})$ is an algebraic integer.

(ii) $\dim U$ divides $|G|$.

Proof. (i) Apply ω_U to Lemma 9.5(ii):

$$\omega_U(\mathcal{K}_i)\,\omega_U(\mathcal{K}_j) = \sum_k a_{ijk}\,\omega_U(\mathcal{K}_k).$$

Fix i, and let w be the column $(\omega_U(\mathcal{K}_k))_k$ and A_i the integer matrix $(a_{ijk})_{j,k}$. The display says $A_i w = \omega_U(\mathcal{K}_i)w$, and $w \neq 0$, as $\omega_U(\mathcal{K}_1) = 1$. Apply Lemma 9.2.

(ii) By Corollary 6.8(i), $\langle \chi_U, \chi_U\rangle_G = 1$. By classes,

$$\begin{aligned} |G| &= \sum_{g\in G} \overline{\chi_U(g)}\,\chi_U(g) \\ &= \sum_k |\mathcal{K}_k|\,\overline{\chi_U(g_k)}\,\chi_U(g_k). \end{aligned}$$

Dividing by $\dim U$, by Lemma 9.5(i),

$$\frac{|G|}{\dim U} = \sum_k \omega_U(\mathcal{K}_k)\,\overline{\chi_U(g_k)}.$$

Integrality of central characters and character values, and closure under sums and products, make the right side integral. The rational-integer criterion (Theorem 9.3) makes the left side an integer. ∎

Lecture 7's count $\sum_U (\dim U)^2 = |G|$ does not give degree divisibility (Example 7.8); integrality does.

Example 9.7 (The 3-cycles of S_3). Let $\mathcal{K}_2$ be the transpositions and $\mathcal{K}_3$ the 3-cycles. Expanding,

$$\begin{aligned} \widehat{\mathcal{K}_3}^2 &= \big((1\,2\,3) + (1\,3\,2)\big)^2 \\ &= (1\,3\,2) + 2e + (1\,2\,3) = 2e + \widehat{\mathcal{K}_3}. \end{aligned}$$

Each of the six products of a 3-cycle and a transposition is odd, hence a transposition. Their sum is central, so the three transpositions have equal coefficients, each 2. Thus

$$\widehat{\mathcal{K}_3}e = \widehat{\mathcal{K}_3}, \qquad \widehat{\mathcal{K}_3}\widehat{\mathcal{K}_2} = 2\widehat{\mathcal{K}_2},$$

and

$$M = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 2 & 0 \\ 2 & 0 & 1 \end{pmatrix}, \qquad \det(t\mathbb{1} - M) = (t-2)^2(t+1).$$

Here $M = (a_{3jk})_{j,k}$ has rows and columns ordered by $\mathcal{K}_1, \mathcal{K}_2, \mathcal{K}_3$; it is the matrix A_i of the proof for $i = 3$. The central-character formula and Example 6.4 give $\omega_E(\mathcal{K}_2) = 0$ and $\omega_E(\mathcal{K}_3) = -1$. Thus the vector $w = (\omega_E(\mathcal{K}_k))_k$ is $(1, 0, -1)^{\mathsf{T}}$, and

$$Mw = (-1, 0, 1)^{\mathsf{T}} = -w.$$

Without the factor $|\mathcal{K}|$, $\chi_E((1\,2\,3))/\dim E = -\tfrac{1}{2}$ is not integral.

Example 9.8 (Guard: integrality needs a representation)**.** On S_3, let φ take the values $1, 1, -1$ on e, the transpositions and the 3-cycles. It is integer-valued, of norm

$$\langle \varphi, \varphi \rangle_{S_3} = \frac{1}{6}(1 + 3 + 2) = 1.$$

But $\langle 1_G, \varphi \rangle_{S_3} = \frac{1}{3}$, so φ is not an integral combination of characters, and Example 6.11 does not apply. For $\varphi = \chi_U$, Lemma 9.5(i) would give $\omega_U(\mathcal{K}_3) = -2$, not an eigenvalue of M. Integrality comes from the representation.

9.3 Averages of roots of unity

For an irreducible representation U and an element $g \in G$, the ratio $\chi_U(g)/\dim U$ is an average of roots of unity. We determine when this average can be an algebraic integer.

Definition 9.9 (Minimal polynomial, conjugates)**.** Let $\alpha \in \mathbb{C}$ be a root of a non-zero polynomial with rational coefficients. Its *minimal polynomial* p is a monic polynomial with rational coefficients, of least degree, with $p(\alpha) = 0$. The roots of p are the *conjugates* of α. Here p is a polynomial; in Burnside's theorem below p is a prime.

Lemma 9.10 (Minimal polynomials)**.** *Let p be the minimal polynomial of α.*

(i) p divides every polynomial f with rational coefficients and $f(\alpha) = 0$. So p is unique.

(ii) If $\alpha \neq 0$, then $p(0) \neq 0$.

(iii) If α is an algebraic integer, p has integer coefficients.

Proof. (i) Divide with remainder: $f = qp + s$ with $\deg s < \deg p$. Then $s(\alpha) = 0$, so $s = 0$ by minimality. Two minimal polynomials divide each other and are monic, so they are equal.

(ii) If $p(0) = 0$, then $p(t) = t\,q(t)$ with q monic, and $q(\alpha) = 0$ as $\alpha \neq 0$, against minimality.

(iii) *The roots.* Let $h(\alpha) = 0$ as in Definition 9.1. By the divisibility property of the minimal polynomial, p divides h, so each root β_k of p is a root of h, an algebraic integer.

The coefficients. Listing the roots with multiplicities, over $\mathbb{C}$ we have $p(t) = \prod_k (t - \beta_k)$, so each coefficient of p is, up to sign, a sum of products of the β_k, an algebraic integer by Theorem 9.3(i).

Integers. They are rational, so integers by Theorem 9.3(ii). ∎

For each integer $m \geq 1$ put $\zeta = \exp(2\pi \mathrm{i}/m)$ and define the *mth cyclotomic polynomial* by

$$\Phi_m(t) = \prod_{0 \leq j < m,\ \gcd(j,m)=1} (t - \zeta^j).$$

For $m = 1$ the only index is $j = 0$. Each m-th root has order ℓ for exactly one $\ell \mid m$, and the roots of order ℓ are the roots of Φ_ℓ. Since $t^m - 1$ has m distinct roots,

$$t^m - 1 = \prod_{\ell \mid m} \Phi_\ell(t).$$

The first three cyclotomic polynomials are

$$\Phi_1(t) = t - 1, \qquad \Phi_2(t) = t + 1, \qquad \Phi_3(t) = t^2 + t + 1.$$

By induction on m, the product of the Φ_ℓ for proper divisors ℓ of m has rational coefficients and is monic; for $m = 1$ the product is empty and equals 1. Polynomial division over $\mathbb{Q}$ gives a rational quotient; uniqueness of the quotient over $\mathbb{C}$ shows that it is Φ_m. Thus every Φ_m has rational coefficients.

Lemma 9.11 (Averages of roots of unity). *Let $\zeta_1, \dots, \zeta_d$ be roots of unity, $d \geq 1$, and $\alpha = (\zeta_1 + \cdots + \zeta_d)/d$. If α is an algebraic integer, then $\alpha = 0$ or $\zeta_1 = \cdots = \zeta_d$.*

Proof. Suppose the ζ_i are not all equal; we show $\alpha = 0$.

The averages α_j. Choose m with every $\zeta_i^m = 1$. As the ζ_i are not all equal, $m \geq 2$. Put $\zeta = \exp(2\pi\mathrm{i}/m)$, write $\zeta_i = \zeta^{k_i}$ with $0 \leq k_i < m$, and let $f(t) = t^{k_1} + \cdots + t^{k_d}$. For j prime to m, let

$$\alpha_j = \frac{f(\zeta^j)}{d} = \frac{\zeta_1^j + \cdots + \zeta_d^j}{d},$$

so $\alpha_1 = \alpha$. Write $uj + vm = 1$ (Bézout). For an m-th root of unity x,

$$x = x^{uj+vm} = (x^j)^u,$$

so $x \mapsto x^j$ is injective on them, and the ζ_i^j are not all equal. So the triangle inequality is strict, as in Proposition 6.17(i): $|\alpha_j| < 1$.

A rational polynomial. Let C be the companion matrix of Φ_m, of size $n = \deg \Phi_m$, with rational entries. Each root θ of Φ_m gives $Cv_\theta = \theta v_\theta$ with $v_\theta = (1, \theta, \dots, \theta^{n-1})^{\mathsf{T}}$; the eigenvalues are distinct, so the v_θ are a basis, and $f(C)v_\theta = f(\theta)v_\theta$. So the rational matrix $f(C)/d$ has the eigenvalues α_j, and

$$F(t) = \det\bigl(t\mathbb{1} - f(C)/d\bigr) = \prod_{0 \leq j < m,\ \gcd(j,m)=1} (t - \alpha_j)$$

has rational coefficients.

The conjugates. $F(\alpha) = 0$, so the minimal polynomial p of α divides F (Lemma 9.10(i)): every conjugate of α is some α_j.

The constant term. By Lemma 9.10(iii), p has integer coefficients. Its roots $\beta_1, \dots, \beta_s$ have modulus less than 1, so

$$|p(0)| = |\beta_1 \cdots \beta_s| < 1.$$

So the integer $p(0)$ is 0, and $\alpha = 0$ by Lemma 9.10(ii). ∎

Example 9.12 (An average and its minimal polynomial). Take $\alpha = (1 + \mathrm{i})/2$, an average of distinct fourth roots of unity. For $m = 4$ and $f(t) = 1 + t$, $\Phi_4(t) = t^2 + 1$, and the construction above gives

$$C = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad \frac{f(C)}{2} = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{pmatrix}, \qquad F(t) = t^2 - t + \tfrac{1}{2}.$$

The roots are $(1 \pm \mathrm{i})/2$, each of modulus $1/\sqrt{2}$. The discriminant of F is -1, so F is irreducible over $\mathbb{Q}$. The minimal polynomial of α divides F (Lemma 9.10(i)). Since F is irreducible, the monic minimal polynomial is F. Its constant coefficient $\frac{1}{2}$ is not an integer, so α is not an algebraic integer by Lemma 9.10(iii). The non-integral coefficient would not suffice without first knowing that F is minimal.

9.4 Burnside's theorem

Theorem 9.13 (Burnside).

(i) Let p be prime and $k \geq 1$. If a conjugacy class of G has p^k elements, then G has a normal subgroup N with $\mathbf{1} \neq N \neq G$; so G is not simple.

(ii) Let $p \neq q$ be primes and $a, b \geq 0$. Every group of order $p^a q^b$ is solvable.

Proof. (i) Let $g \in \mathcal{K}$, with $|\mathcal{K}| = p^k > 1$. Then $g \notin Z(G)$; in particular $g \neq e$.

A character. By column orthogonality (Theorem 7.11(iii)) for e and $g \notin [e]$,

$$0 = \sum_{U \in \mathrm{Irr}(G)} \overline{\chi_U(e)}\, \chi_U(g) = 1 + \sum_{U \neq \mathbf{1}} \dim U\, \chi_U(g).$$

If p divided $\dim U$ for every $U \neq \mathbf{1}$ with $\chi_U(g) \neq 0$, then

$$-\frac{1}{p} = \sum_{U \neq \mathbf{1},\ \chi_U(g) \neq 0} \frac{\dim U}{p}\, \chi_U(g)$$

would be an algebraic integer by closure, contradicting the rational-integer criterion (Theorem 9.3). So some $U \in \mathrm{Irr}(G)$, $U \neq \mathbf{1}$, has $p \nmid \dim U$ and $\chi_U(g) \neq 0$.

A scalar. Then $|\mathcal{K}|$ and $\dim U$ are coprime, so $ap^k + b \dim U = 1$ for some integers a and b, and

$$\begin{aligned} \frac{\chi_U(g)}{\dim U} &= a\, \frac{p^k \chi_U(g)}{\dim U} + b\, \chi_U(g) \\ &= a\, \omega_U(\mathcal{K}) + b\, \chi_U(g), \end{aligned}$$

an algebraic integer (Theorem 9.6(i), Theorem 9.3). It is the average of the $\dim U$ eigenvalues of $\rho_U(g)$, roots of unity (Corollary 5.17(ii)), and it is not 0. By Lemma 9.11 the eigenvalues are equal, so $|\chi_U(g)| = \dim U$, and $\rho_U(g) = c\mathbb{1}$ by Proposition 6.17(i).

A normal subgroup. $N = \ker \rho_U$ is normal (Proposition 6.17(ii)). $N \neq G$: otherwise every line in U is invariant, so the irreducible U is a line with trivial action, $U \cong \mathbf{1}$. $N \neq \mathbf{1}$: otherwise ρ_U is injective, and as $c\mathbb{1}$ commutes with every $\rho_U(h)$,

$$\rho_U\left(ghg^{-1}h^{-1}\right) = \mathbb{1} \qquad \text{for all } h \in G,$$

so $g \in Z(G)$, which is false.

(ii) By induction on $|G|$. If G is abelian, it is solvable (Definition 4.16). Otherwise it suffices to find $N \trianglelefteq G$ with $\mathbf{1} \neq N \neq G$: then N and G/N have orders $p^{a'}q^{b'}$ less than $|G|$, so they are solvable, and so is G, by Theorem 4.17(iii).

If $Z(G) \neq \mathbf{1}$, take $N = Z(G)$, which is not G. If $Z(G) = \mathbf{1}$, the class equation (Theorem 1.20(ii)) reads

$$|G| = 1 + \sum_{i=1}^{m} [G : C_G(g_i)],$$

each term of the sum exceeding 1 and dividing $p^a q^b$. Since G is nontrivial, some prime divides $|G|$; exchanging the names and their exponents if necessary, let q divide $|G|$. If every term of the sum were divisible by q, the right side would be 1 modulo q, while $|G|$ is 0 modulo q. So a term is prime to q. It divides $p^a q^b$ and exceeds 1, hence is p^k with $k \geq 1$. Part (i) gives N. ∎

Example 9.14 (A prime-power class in D_4). The class of r is $\{r, r^3\}$: conjugation by s sends r to r^{-1}, and conjugation by a power of r fixes it. Thus its size is 2. The one-dimensional character χ is defined using the unique normal forms by $\chi(r^a s^\epsilon) = (-1)^a$ (Example 1.18). The multiplication rule of § 2.3 shows it is a homomorphism: the exponent of r in a product is $a + (-1)^\epsilon b$, congruent to $a + b$ modulo 2. It has degree 1, prime to 2, and $\chi(r) = -1 \neq 0$, so it is a witness of the kind chosen in the proof. Its kernel is

$$\ker \chi = \{e, r^2, s, r^2 s\} = \langle r^2, s \rangle,$$

a nontrivial proper normal subgroup. In this example the witness is already one-dimensional; the proof also handles higher-dimensional witnesses.

9.5 The Frobenius–Schur indicator

An invariant Hermitian inner product is sesquilinear, so it does not by itself give a complex-bilinear form. The indicator distinguishes the invariant bilinear forms on an irreducible representation.

A bilinear form B on a representation V is *invariant* if $B(g\cdot u, g\cdot v) = B(u,v)$ for all g, u, v; it is *symmetric* if $B(u,v) = B(v,u)$, and *alternating* if $B(u,v) = -B(v,u)$. The *Frobenius–Schur indicator* of V is

$$\nu(V) = \frac{1}{|G|}\sum_{g\in G}\chi_V(g^2).$$

Theorem 9.15 (Frobenius–Schur). *Let $U \in \mathrm{Irr}(G)$. The invariant bilinear forms on U form a space of dimension at most 1, and $\nu(U) \in \{1, 0, -1\}$:*

- *$\nu(U) = 1$ if and only if U carries a non-zero invariant symmetric form;*
- *$\nu(U) = -1$ if and only if it carries a non-zero invariant alternating form;*
- *$\nu(U) = 0$ if and only if it carries no non-zero invariant bilinear form, if and only if χ_U is not real-valued.*

Proof. Forms as tensors. Identify $U^*\otimes U^*$ with the bilinear forms on U, $f\otimes f'$ being $(u,v)\mapsto f(u)f'(v)$. Then G acts by $(g\cdot B)(u,v) = B(g^{-1}\cdot u, g^{-1}\cdot v)$ (Definition 6.12), and the swap τ exchanges the arguments. Its $+1$ and -1 eigenspaces are the symmetric and exterior squares, respectively (Definition 6.12). All invariant forms make up $(U^*\otimes U^*)^G$; all symmetric forms make up $\mathrm{Sym}^2(U^*)$, and all alternating forms make up $\Lambda^2(U^*)$. Taking invariants in the splitting of Theorem 6.13 gives

$$(U^*\otimes U^*)^G = \mathrm{Sym}^2(U^*)^G \oplus \Lambda^2(U^*)^G.$$

At most one. By Proposition 5.8(iii) and Theorem 6.13(i), (ii),

$$\begin{aligned}\dim(U^*\otimes U^*)^G &= \frac{1}{|G|}\sum_g \overline{\chi_U(g)}^{\,2}\\ &= \langle \chi_U, \chi_{U^*}\rangle_G .\end{aligned}$$

Here U^* is irreducible, as $\langle\overline{\chi_U},\overline{\chi_U}\rangle_G = 1$. So this dimension is 1 if $\overline{\chi_U}$ equals χ_U, and 0 otherwise (Corollary 6.8(i), (iii)).

The difference. By Theorem 6.13(iii) for U^*, the characters of $\mathrm{Sym}^2(U^*)$ and $\Lambda^2(U^*)$ differ by $\overline{\chi_U(g^2)}$ at g. So

$$\begin{aligned}\dim \mathrm{Sym}^2(U^*)^G - \dim\Lambda^2(U^*)^G &= \frac{1}{|G|}\sum_g \overline{\chi_U(g^2)}\\ &= \overline{\nu(U)}.\end{aligned}$$

The cases. The two dimensions are non-negative integers with sum at most 1. So $\overline{\nu(U)}$, and with it $\nu(U)$, is 1, -1 or 0 according as the non-zero invariant forms are symmetric, alternating, or absent; they are absent exactly when χ_U is not real. ∎

Example 9.16 (Guard: no non-zero invariant bilinear form). When $\nu(U) = 0$, every invariant bilinear form is zero. Let $C_3 = \langle g\rangle$ act on $\mathbb{C}$ by $g\mapsto\zeta = \exp(2\pi\mathrm{i}/3)$. Then

$$\nu(U) = \frac{1}{3}(1+\zeta^2+\zeta^4) = 0,$$

and indeed $B(u,v) = b\,uv$ has $B(g\cdot u, g\cdot v) = \zeta^2 B(u,v)$, so it is invariant only if $b = 0$.

Remark 9.17 (Real and quaternionic · *sketched*)**.** Let U be irreducible. If some basis makes every $\rho_U(g)$ real, then $\nu(U) = 1$: the sum over g of $(D(g)u)^{\mathsf{T}} D(g)v$ is an invariant symmetric form, positive on real vectors. Conversely, $\nu(U) = 1$ makes U *realisable over* $\mathbb{R}$ in this sense, and $\nu(U) = -1$ makes it *quaternionic*: real character, not realisable. *The missing step:* with an invariant inner product (Theorem 5.11(iii)), the invariant form is $\langle Ju, v\rangle$ for an antilinear J commuting with G, with $J^2 = c\mathbb{1}$, $c > 0$, if the form is symmetric, and $J^2 = -c\mathbb{1}$ if alternating. When $J^2 = c\mathbb{1}$ with $c > 0$, the fixed vectors of $J/\sqrt{c}$ span a real form. Over $\mathbb{C}$, Example 5.14's rotation plane splits into the lines of $g \mapsto \mathrm{i}$ and $g \mapsto -\mathrm{i}$, each with $\nu = 0$.

9.6 Worked example

We determine the invariant forms on the irreducible plane of Q_8. Recall

$$Q_8 = \{\pm 1, \pm i, \pm j, \pm k\},$$
$$i^2 = -1, \qquad j^2 = -1, \qquad k^2 = -1, \qquad ijk = -1,$$

with identity $e = 1$; the italic i, j, k are quaternions, and the upright i is the complex unit. Let

$$D(1) = \mathbb{1}, \qquad D(i) = \begin{pmatrix} \mathrm{i} & 0 \\ 0 & -\mathrm{i} \end{pmatrix}, \qquad D(j) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix},$$

and, for $x \in \{1, i, j, k\}$,

$$D(k) = D(i)D(j), \qquad D(-x) = -D(x).$$

They multiply as the quaternions do:

$$D(i)^2 = -\mathbb{1}, \qquad D(j)^2 = -\mathbb{1}, \qquad D(k)^2 = -\mathbb{1},$$
$$D(j)D(k) = D(i), \qquad D(k)D(i) = D(j).$$

The remaining products follow by taking inverses, since $D(x)^{-1} = -D(x)$ for $x = \pm i, \pm j, \pm k$. These are the matrices on the basis $e \otimes w$, $j \otimes w$ of the induced representation U in the $\langle i\rangle \leq Q_8$ case of Example 8.16. The identity acts by $\mathbb{1}$. The character values are 2 at 1, -2 at -1, and 0 at the six remaining elements. Hence

$$\langle \chi_U, \chi_U\rangle_{Q_8} = \frac{1}{8}(4 + 4) = 1,$$

and U is irreducible (Corollary 6.8(iv)). Now ± 1 square to 1, and the six $\pm i$, $\pm j$, $\pm k$ square to -1:

$$\nu(U) = \frac{1}{8}\big(2\chi_U(1) + 6\chi_U(-1)\big) = \frac{1}{8}(4 - 12) = -1.$$

So U carries an invariant alternating form, $B(u, v) = u_1v_2 - u_2v_1$, as each $D(x)$ has determinant 1; and no non-zero symmetric one.

Self-check. Which form does the plane of D_4 carry? *Answer:* with Lecture 5's matrices (§ 5.5), the traces give $\chi(e) = 2$ and $\chi(r^2) = -2$. The elements e, r^2 and the four reflections square to e, and r, r^3 to r^2:

$$\nu = \frac{1}{8}\big(6 \cdot 2 + 2 \cdot (-2)\big) = 1,$$

with the symmetric form $u_1v_2 + u_2v_1$. D_4 and Q_8 have the same character table (Sheet 6); computing the indicator also requires the class of each square g^2. The character table alone does not specify those classes.

Summary.

- *The main results:* character values and central characters are algebraic integers, so every irreducible degree divides $|G|$ (Theorems 9.3 and 9.6). An algebraic-integer average of roots of unity is zero or all the roots averaged are equal (Lemma 9.11). A class of prime-power size greater than one gives a nontrivial proper normal subgroup; groups of order $p^a q^b$, with distinct primes p, q and $a, b \geq 0$, are solvable (Theorem 9.13). For an irreducible complex representation, the Frobenius–Schur indicator is 1 for a nonzero invariant symmetric form, -1 for an alternating one, and 0 when neither exists (Theorem 9.15).
- *Handed on:* the indicator to the *Category Theory* course; the integer-valued norm-one class function that is not a character (Example 9.8) to Lecture 13.
- *Next:* Lecture 10 begins the representations of S_n.

References

Serre, *Linear Representations of Finite Groups*: §§6.3–6.5, pp. 50–53, with Exercises 6.8–6.10; Burnside's theorem, Exercise 8.6, p. 65; §13.2, pp. 106–109. James and Liebeck, *Representations and Characters of Groups*: Ch. 22, pp. 244–250; the indicator, Thm 23.16, pp. 275–277; Burnside's theorem, Ch. 31, pp. 362–365. Etingof et al., *Introduction to Representation Theory*: §§5.1–5.4, pp. 92–100.[1]

[1] All three put the antilinear slot second and write $\chi(1)$ for $\dim U$.